V4.2 Toy Model 1D-C2

Directional Delay + Vectorial Connectivity

Purpose:
Test Catalin's second correction: delay should not remain purely scalar.

Cost rule:
c_ij = c0_ij * [1 + lambda0 R_lambda + lambda_r R_lambda (e_hat dot r_hat)^2]
/ [1 + mu R_mu (e_hat dot g_hat)^2]

Formula meaning:
lambda0 = minor scalar/general delay sensitivity.
lambda_r = radial directional delay sensitivity.
mu = vectorial connectivity sensitivity.
R_lambda = wide delay field, rho_lambda = 4.0.
R_mu = tight connectivity field, rho_mu = 2.0.
(e_hat dot r_hat)^2 = radial-alignment penalty for delay.
(e_hat dot g_hat)^2 = radial-alignment shortcut for connectivity.
The squared dot products preserve symmetry: inward and outward radial movement are treated equally.

Diagnostics:
L = scout delay relative to flat baseline.
F_center > 0 = off-axis paths moved inward near the centre.
Detour > 0 = off-axis paths moved outward near the centre.
Inner paths = number of bundle paths entering the tight connectivity radius.
Central crossings = paths that pass within |y| <= 1 near x=0.

Guardrail:
This is not General Relativity and not gravitational lensing. It is a controlled graph test
of whether directional delay plus vectorial connectivity can produce more lensing-like path behaviour.

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Numerical summary:

C2_flat_control: λ0=0.0, λr=0.0, μ=0.0, L=0.000, avgD=0.000, F=0.000, detour=0.000, inner=3/9, central=1/9
C2_lambda0_only: λ0=0.5, λr=0.0, μ=0.0, L=4.952, avgD=2.695, F=0.000, detour=0.000, inner=3/9, central=1/9
C2_radial_delay_only: λ0=0.5, λr=1.0, μ=0.0, L=7.874, avgD=3.728, F=-1.750, detour=1.750, inner=0/9, central=0/9
C2_connect_only: λ0=0.0, λr=0.0, μ=3.0, L=-5.369, avgD=-2.040, F=3.000, detour=-3.000, inner=7/9, central=7/9
C2_balanced_lamr1_mu1: λ0=0.5, λr=1.0, μ=1.0, L=7.843, avgD=3.722, F=-1.750, detour=1.750, inner=0/9, central=0/9
C2_focus_lamr1_mu2: λ0=0.5, λr=1.0, μ=2.0, L=4.667, avgD=3.367, F=-1.750, detour=1.750, inner=1/9, central=1/9
C2_focus_lamr2_mu3: λ0=0.5, λr=2.0, μ=3.0, L=8.086, avgD=4.042, F=-2.250, detour=2.250, inner=1/9, central=1/9
C2_delay_lamr3_mu1: λ0=0.5, λr=3.0, μ=1.0, L=8.935, avgD=4.357, F=-2.750, detour=2.750, inner=0/9, central=0/9
C2_strong_lamr3_mu5: λ0=0.5, λr=3.0, μ=5.0, L=8.913, avgD=4.353, F=-2.750, detour=2.750, inner=0/9, central=0/9
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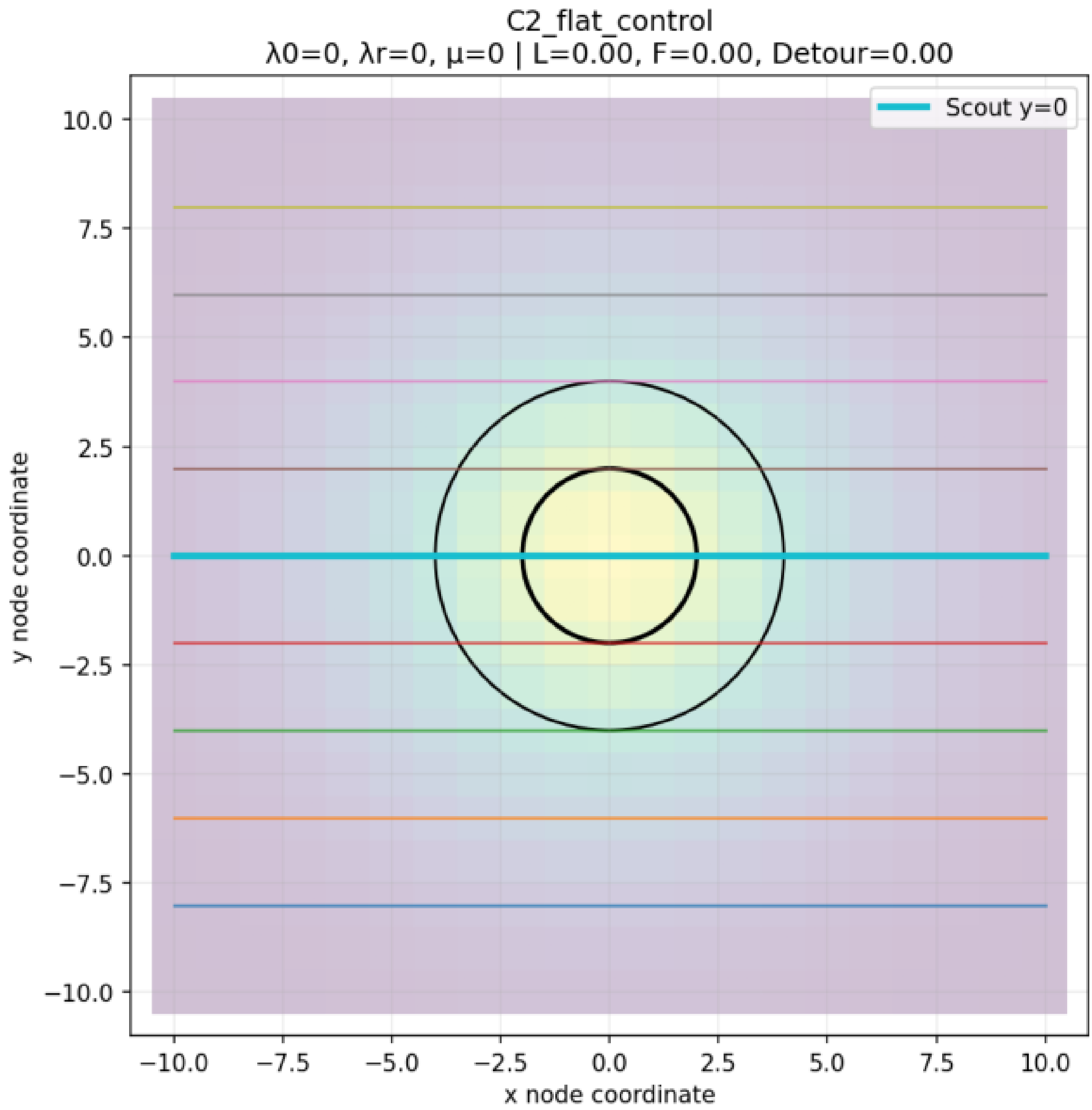
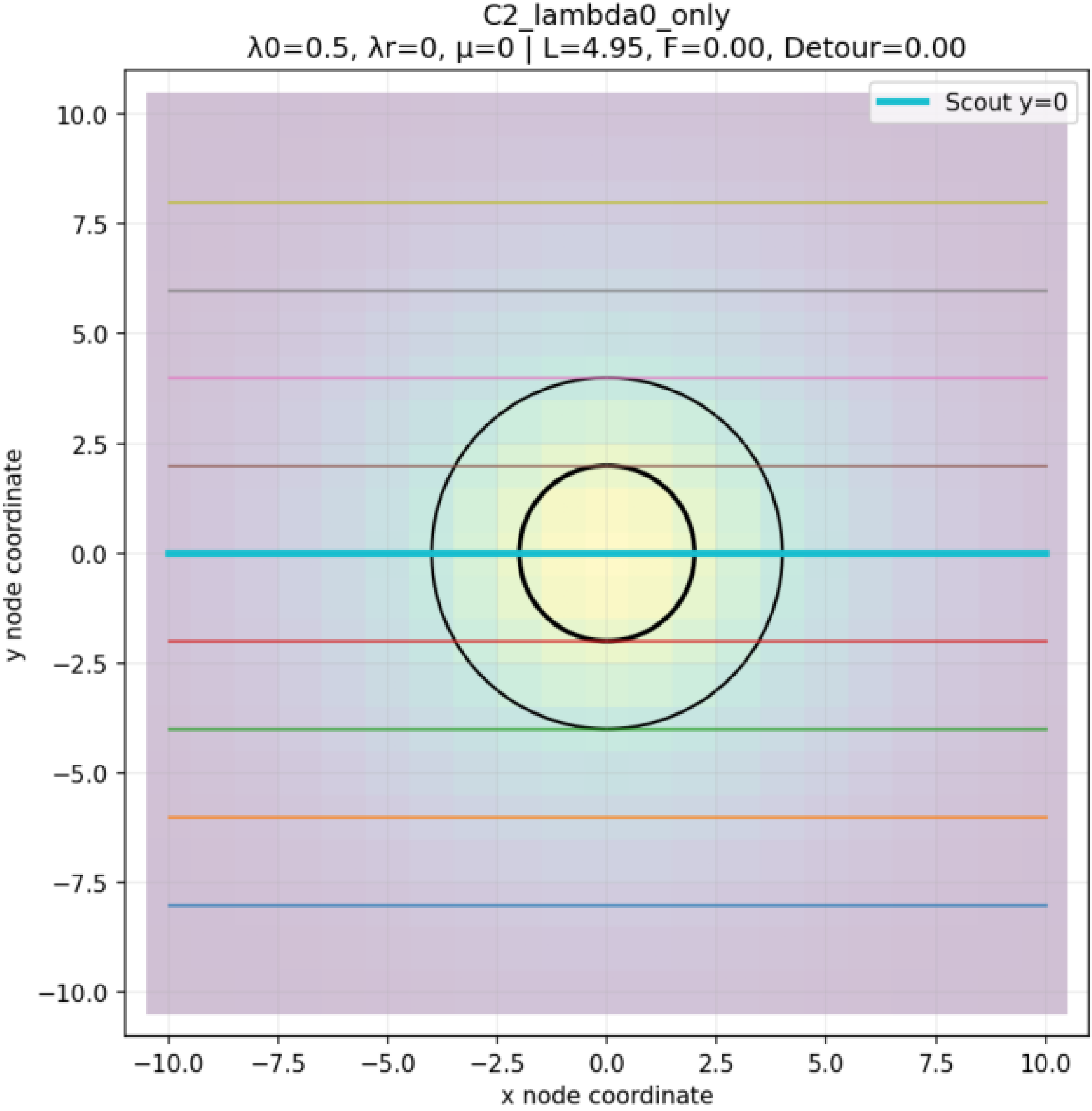
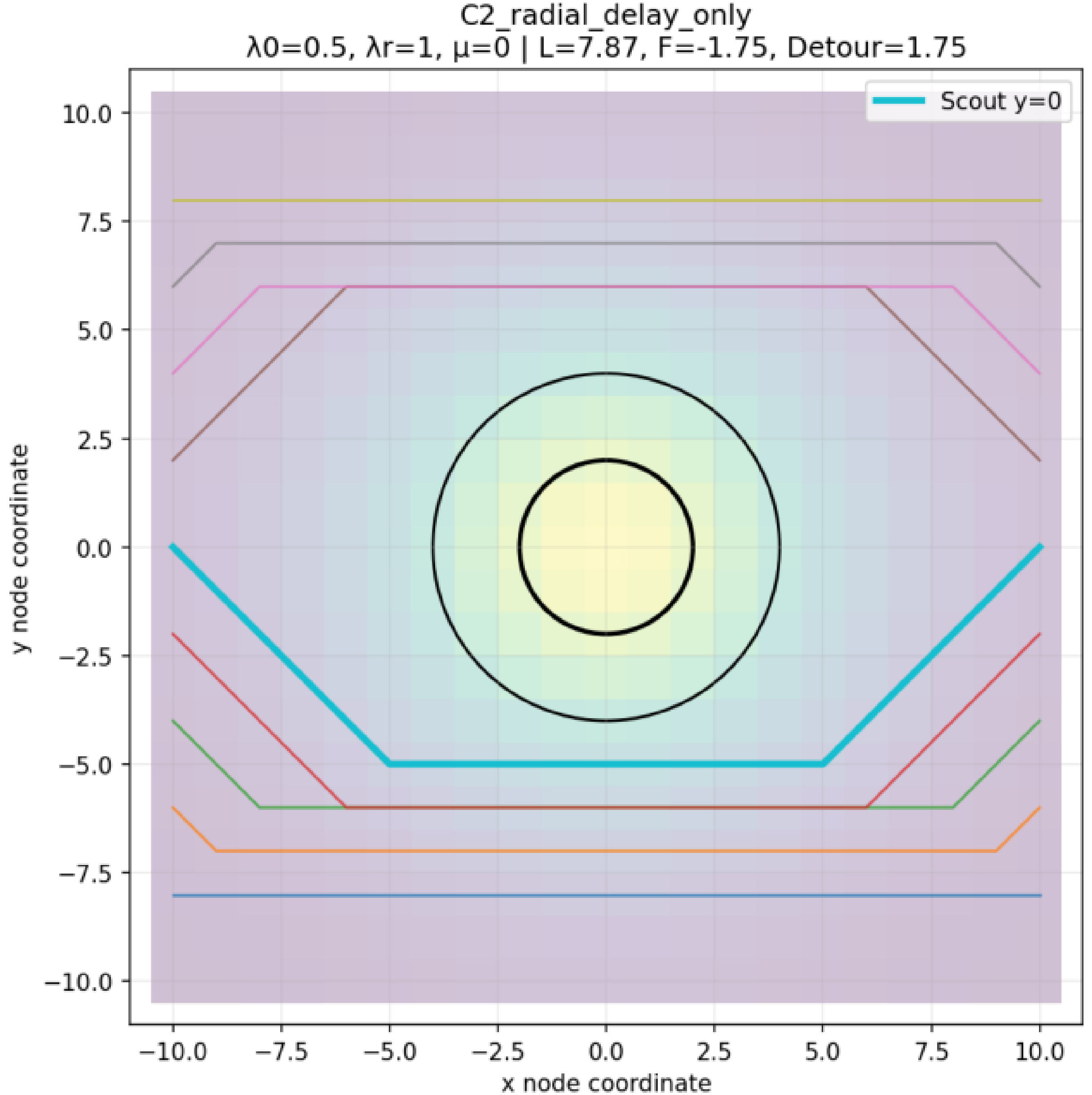
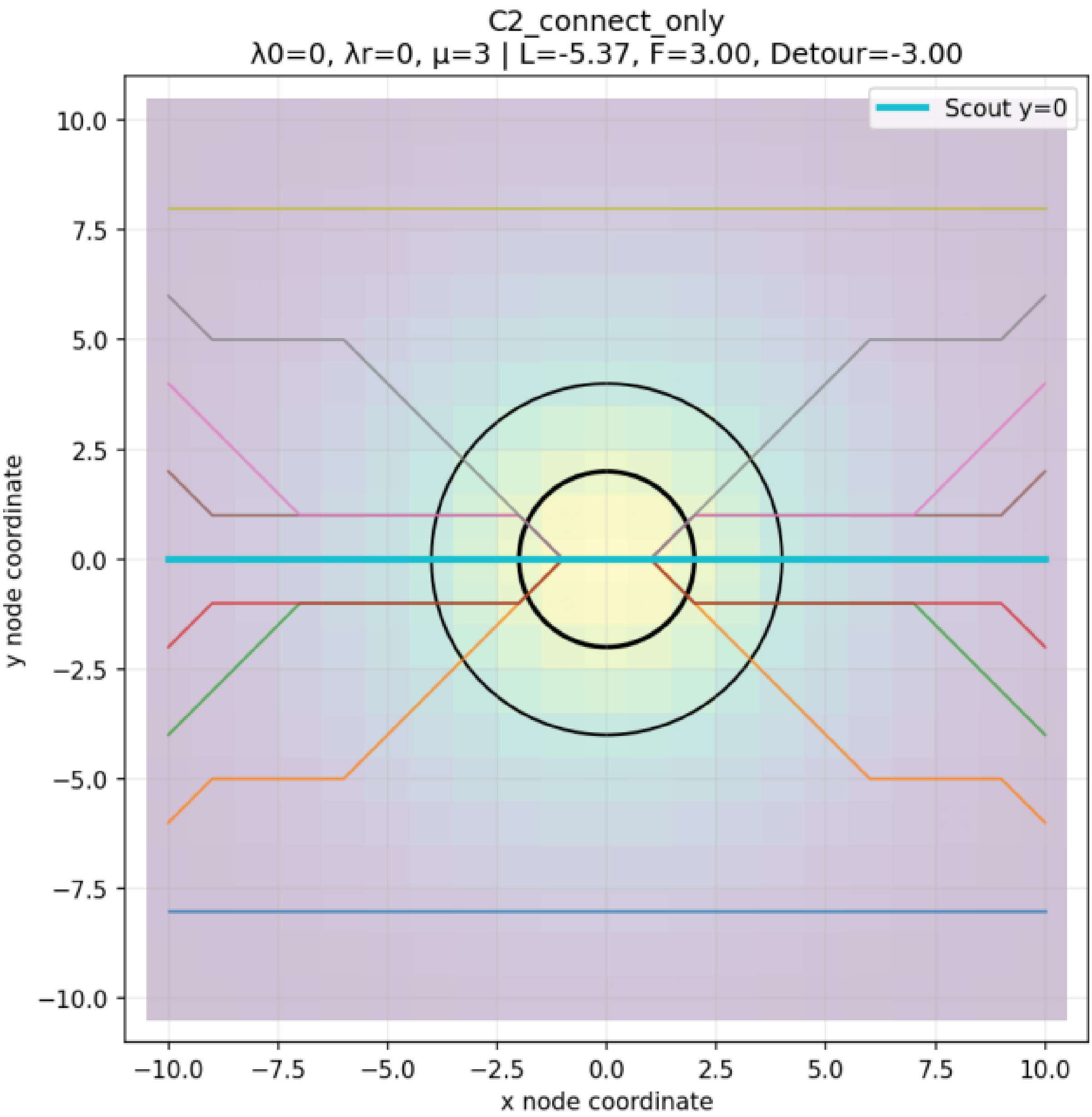

C2_flat_control
λ0=0, λr=0, μ=0 | L=0.00, F=0.00, Detour=0.00
Scout y=0
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
y node coordinate
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x node coordinate

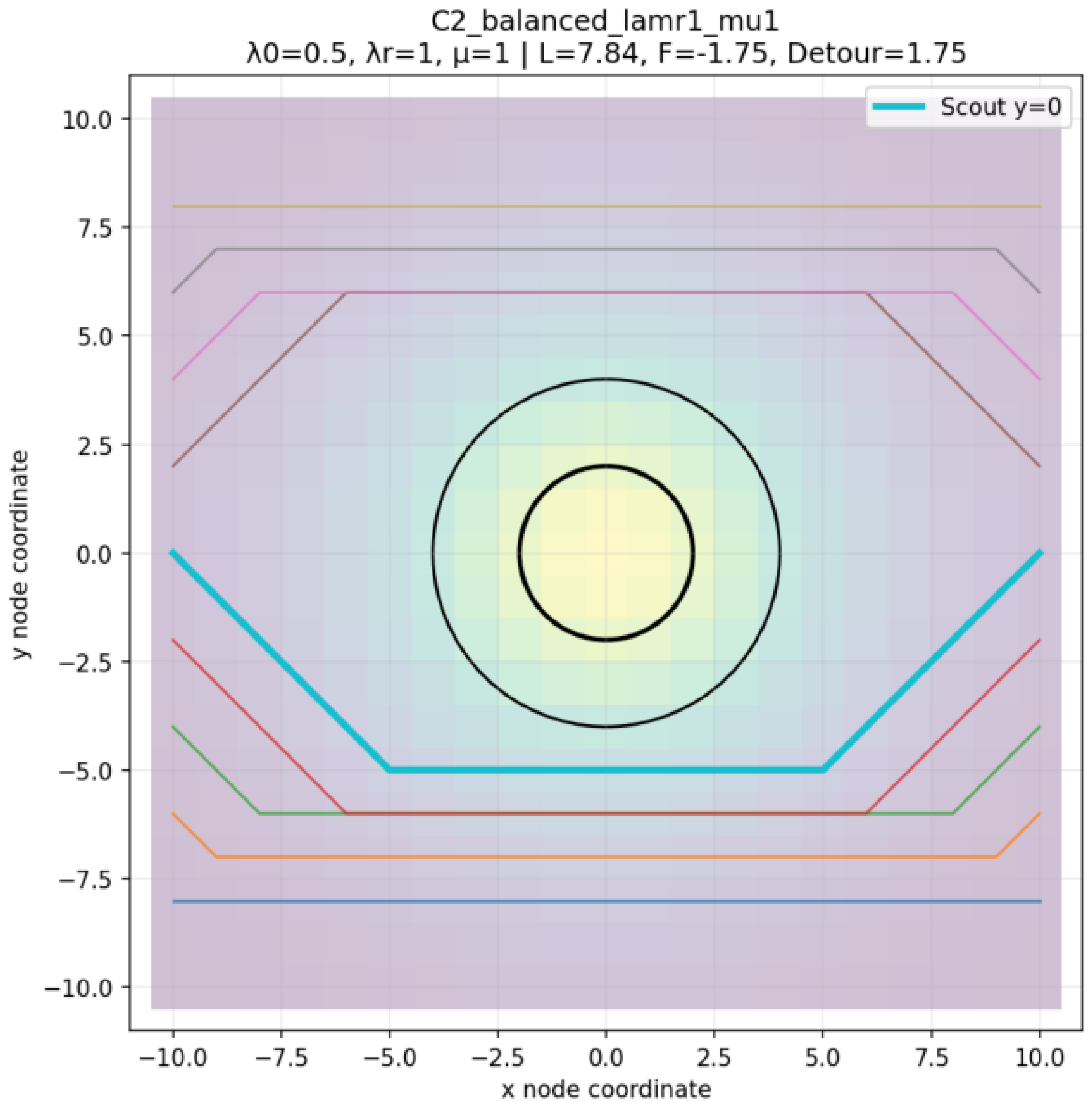


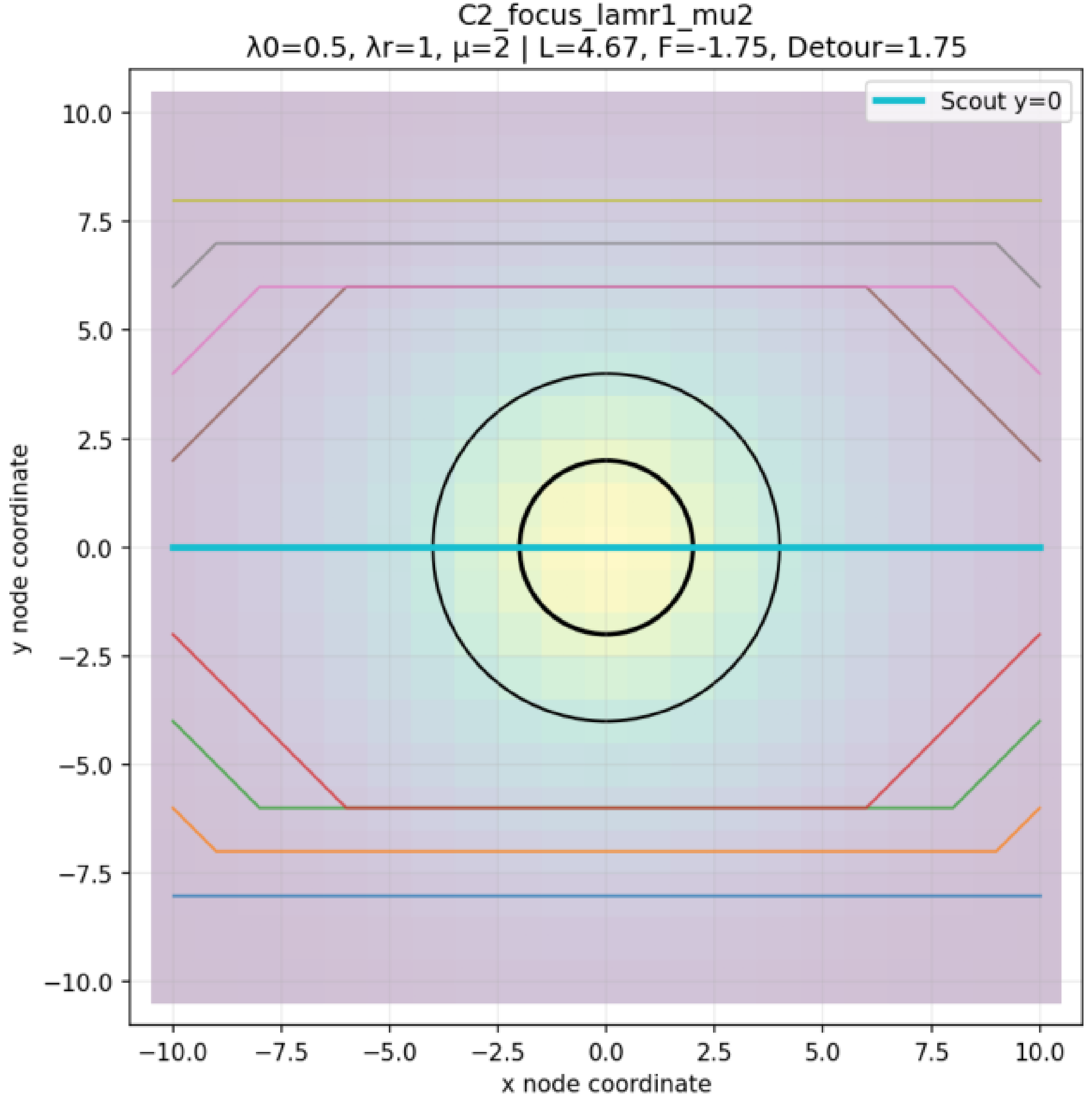
C2_lambda0_only
λ0=0.5, λr=0, μ=0 | L=4.95, F=0.00, Detour=0.00
Scout y=0
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
y node coordinate
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x node coordinate


C2_radial_delay_only
λ0=0.5, λr=1, μ=0 | L=7.87, F=-1.75, Detour=1.75
Scout y=0
y node coordinate
x node coordinate
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0

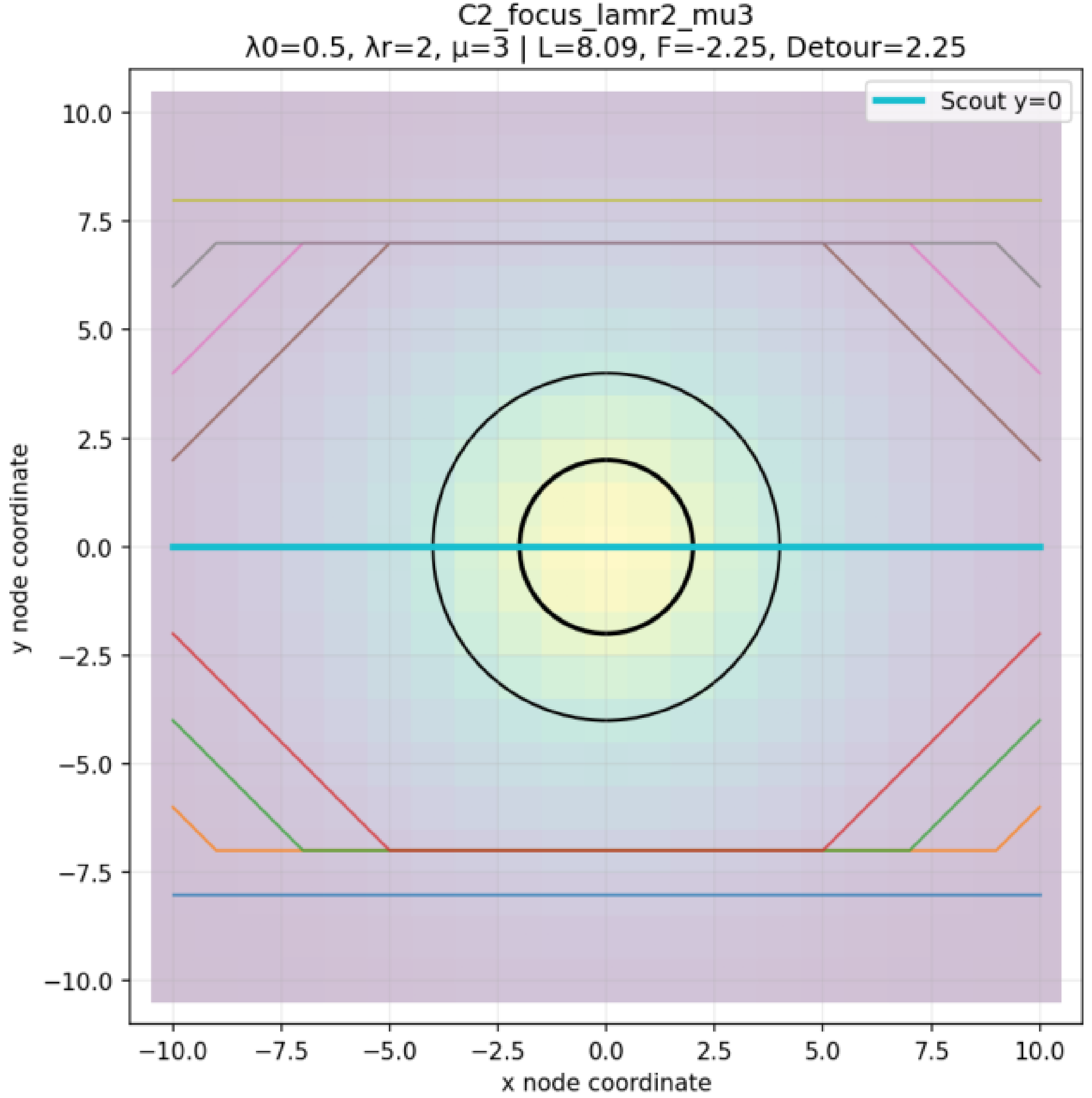


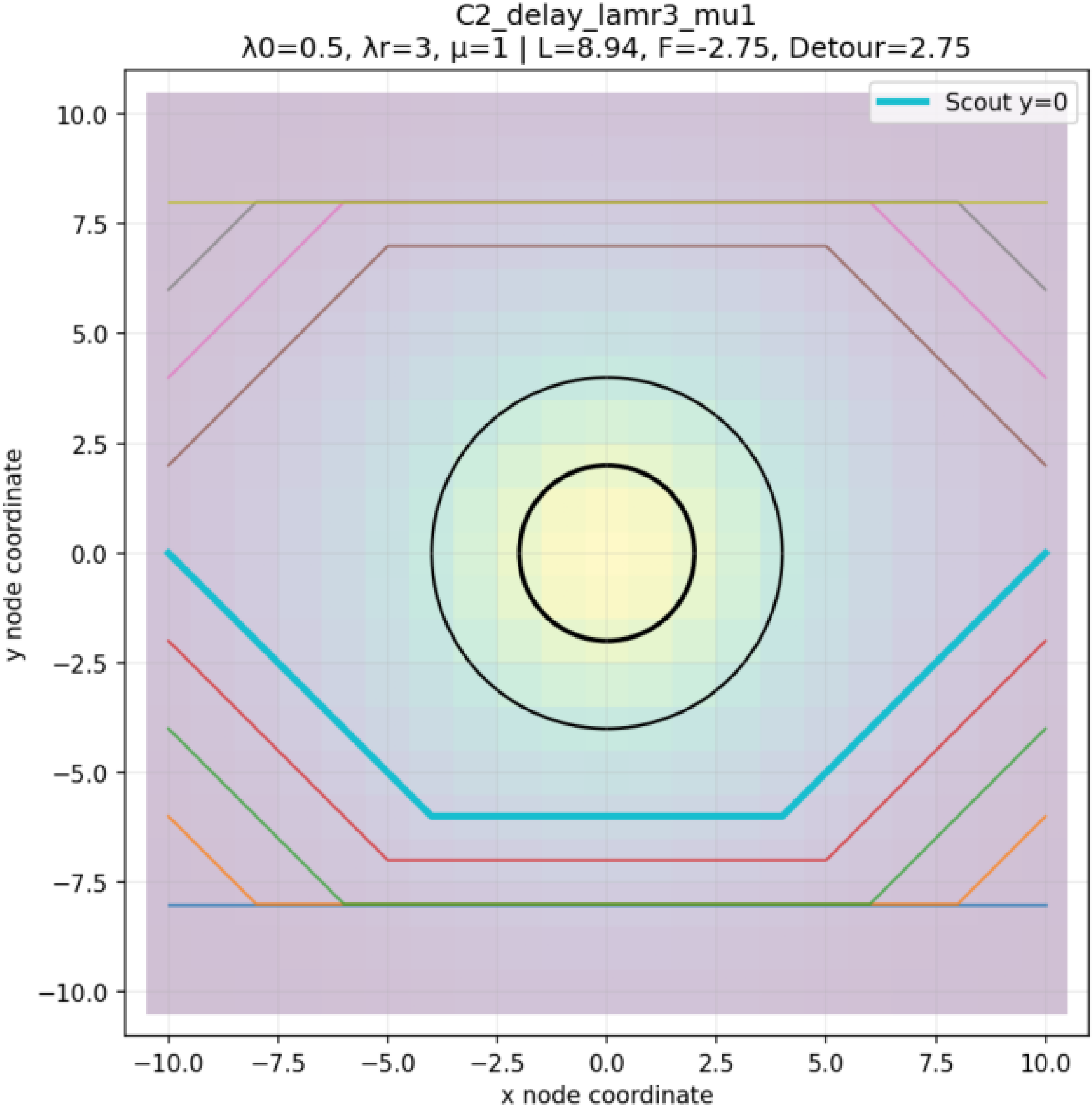
C2_connect_only
λ0=0, λr=0, μ=3 | L=-5.37, F=3.00, Detour=-3.00
Scout y=0
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
y node coordinate
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x node coordinate

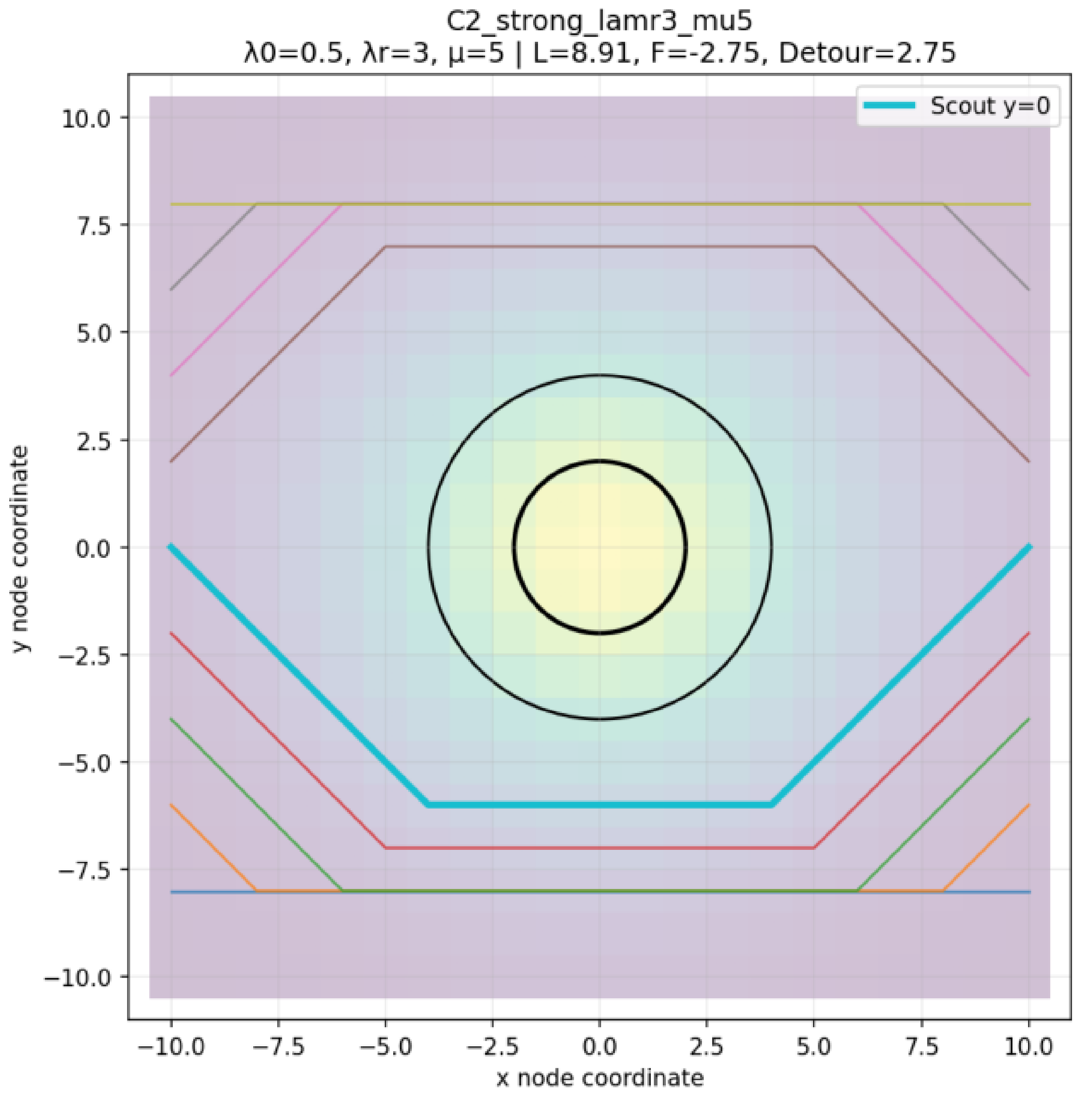

C2_balanced_lamr1_mu1
λ0=0.5, λr=1, μ=1 | L=7.84, F=-1.75, Detour=1.75
Scout y=0
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
y node coordinate
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x node coordinate



C2_focus_lamr1_mu2
λ0=0.5, λr=1, μ=2 | L=4.67, F=-1.75, Detour=1.75
Scout y=0
y node coordinate
x node coordinate
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0


C2_focus_lamr2_mu3
λ0=0.5, λr=2, μ=3 | L=8.09, F=-2.25, Detour=2.25
Scout y=0
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
y node coordinate
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x node coordinate


C2_delay_lamr3_mu1
λ0=0.5, λr=3, μ=1 | L=8.94, F=-2.75, Detour=2.75
Scout y=0
y node coordinate
x node coordinate
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0



C2_strong_lamr3_mu5
λ0=0.5, λr=3, μ=5 | L=8.91, F=-2.75, Detour=2.75
Scout y=0
10.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
y node coordinate
−10.0
−7.5
−5.0
−2.5
0.0
2.5
5.0
7.5
10.0
x node coordinate